Embedded Isolates Define a Target-Specific Temporal Response Law in Cellular Automata

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(Dated: 29 April 2026)

A single structural feature of a cellular automaton state—the count of *embedded isolated cells* (alive cells with no 4-connected live neighbour but at least one diagonal live neighbour)—predicts future fine-scale component change in Conway's Game of Life and HighLife with a remarkably stable negative coefficient. We report four interlocking empirical regularities. First, a *target-specific selection principle*: after residualising on density, iso_count adds $\Delta R^2 = 0.0012$ for the fine-net component target (GoL: PASS; global: PASS) while the iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), confirming the relation is not a density artefact. Second, a *temporal response law*: the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative and stable—ranging from -0.798 at $k = 1$ to -0.709 at $k = 200$, CV $\lesssim 0.17$—across all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$, 2 rules $\times$ 2 sizes $\times$ 4 density levels = 16 conditions). Third, a *local-context mechanism*: the one-step carrier is local component-context loss (CV $R^2 = 0.538$), not simple cell death, and this mechanism transfers across all leave-one-out axes (density, size, rule, condition) after condition-standardisation ($R_z^2 = 0.544$; frac R^2-positive = 1.00). Fourth, a *two-layer amplitude structure*: the standardised mechanism direction is transferable, while the raw amplitude is predictable from grid size L and density ρ with leave-one-out $R^2 = 0.977$, giving the response a clean decomposition. Using only $t = 0$ iso_count (no fate features), the full multi-horizon response is recovered (mean $R^2 = 0.241$ over all horizons; shuffle ≈ 0.0008), establishing iso_count as the *sufficient non-leaky prestate object summary.* Finally, fine-net horizon tasks are rank-1 coherent in whitened description space (mean pairwise $|\cos| = 0.999$; rank-1 energy = 0.999), while heterogeneous prediction targets are not ($|\cos| = 0.427$; rank-1 energy = 0.532), providing a formal bridge to observer-dependent description theory.

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response law, mechanism transfer, observer-dependent description

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I. INTRODUCTION

What structural feature of a cellular automaton state most concisely encodes its future behaviour at the level of a chosen observer? This question sits at the intersection of computational mechanics, coarse-graining theory, and the emerging framework of observer-dependent description[1–3]. Conway's Game of Life (GoL)[4] and HighLife provide unusually clean test beds: the dynamics are deterministic and reproducible, multiple natural coarse-graining conventions exist (component counts, block densities, entropy measures), and systematic numerical experiments are computationally affordable. Earlier work established the edge-of-chaos properties[5,6] and statistical complexity[7] of Life-like rules, and identified the role of coherent structures[8] and computational irreducibility[9] in deterministic CAs.

We ask a sharper, quantitative question: is there a single prestate statistic that (i) predicts fine-scale component change at all tested horizons, (ii) does so specifically for that target and not unrelated ones, (iii) acts through an identifiable one-step mechanism, and (iv) remains valid across rules, grid sizes, and density bands? The answer is yes, and the statistic is the count of *embedded isolated cells* (hereafter iso_count): alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour.

Why should embedded isolates be structurally meaningful? An alive cell that is 4-disconnected but diagonally attached sits precisely at the boundary between topological isolation and connectivity. Its four orthogonal neighbours are all dead, making it a fragile local singularity; yet it is diagonally tethered, so small perturbations can either fully isolate or reconnect it. These cells are therefore positioned at the most unstable locus of the local component topology. When such a cell dies—or when diagonal neighbours die around it—the local connectivity changes in ways that propagate into the fine-net component count. The iso_count statistic measures how many such "fragile bridge nodes" are present at time $t = 0$, and this paper shows that this count is sufficient to predict the aggregate fine-net change at all tested horizons.

This paper establishes five interlocking results in a single chain:

1. *Target-specific selection*: iso_count uniquely improves prediction for the fine-net target, not for density, entropy, or coarse-block targets.

2. *Non-leaky prestate summary*: using only $t = 0$ iso_count, without any fate information,

recovers the full residual $R^2(k)$ at every horizon.

3. *Temporal response law*: the slope $\hat{\beta}_{\text{iso}}(k)$ is negative and stable from $k = 1$ to $k = 200$ across all 112 condition-horizon combinations.

4. *Mechanism identity and transfer*: the one-step carrier is local component-context loss, not cell death, and this mechanism transfers across all leave-one-out generalisation axes.

5. *Task-direction coherence*: fine-net horizon tasks form a rank-1 coherent family in description space, providing a formal geometric explanation for why a single statistic serves all horizons.

Figure 1 gives the entry-point overview: the structural definition of an embedded isolate (panel a) and the target-specific ΔR^2 pattern that selects it (panel b).

II. DEFINITIONS AND PROTOCOL

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a). We denote the 4-connected component count at time t as $C(t)$ and define the fine-net change at horizon k as $\Delta C_k = C(t + k) - C(t)$.

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons $\times$ 16 conditions = 112 condition-horizon slope estimates.

The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares with $N_{\text{live}}(0)$ (live-cell count, proportional to density) as a nuisance covariate, so that iso_count is evaluated for its density-

orthogonal contribution. R^2 values for mechanism features are cross-validated (5-fold). We do not rerun experiments; all statistics are drawn from the fixed set of CSVs in `outputs/`.

III. RESULT 1 — TARGET-SPECIFIC SELECTION

The central diagnostic is a selection-principle battery (Fig. 1b): for each candidate predictor (iso_count, coarse block-variance, block entropy, iso-shuffle null, target-shuffle null) we measure $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$ on seven prediction targets.

For iso_count on the fine-net target, the global pooled result is $\Delta R^2 = 0.0012$ (GoL: PASS; global: PASS). The iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$ for both), confirming the signal is not an artefact of the density-count correlation. Crucially, iso_count adds almost nothing for the density, block-variance, block-entropy, or coarse-component targets: the ΔR^2 uplift is concentrated on the fine-net target only. The fine-net iso slope is -0.1054 (negative), consistent across all conditions.

This target specificity rules out explanations based purely on density or total alive-cell geometry. An iso-shuffled predictor of the same cardinality provides effectively zero incremental R^2, establishing that the spatial arrangement encoded in iso_count—not just its count—carries the signal.

IV. RESULT 2 — NON-LEAKY PRESTATE SUMMARY

A natural concern is whether iso_count at $t = 0$ is merely a proxy for information that requires knowing the initial fate of individual cells. We test this with a prestate-only horizon recovery: using *only* iso_count at $t = 0$—no one-step fate features, no local-window information, no knowledge of any transition at $t \to t + 1$—we ask how well the multi-horizon residual $R^2(k)$ is recovered.

Figure 2 shows the result: the prestate-only model achieves mean $R^2 = 0.241$ over all horizons $k \in \{1, \ldots, 200\}$, minimum $R^2 = 0.175$, versus shuffle $R^2 \approx 0.0008$. Adding the full class-count breakdown does not substantially improve over iso_count alone. The full temporal response is encoded in the $t = 0$ prestate; no information leaks in from later timesteps.

This non-leakiness result has a structural interpretation: the embedded isolate count at

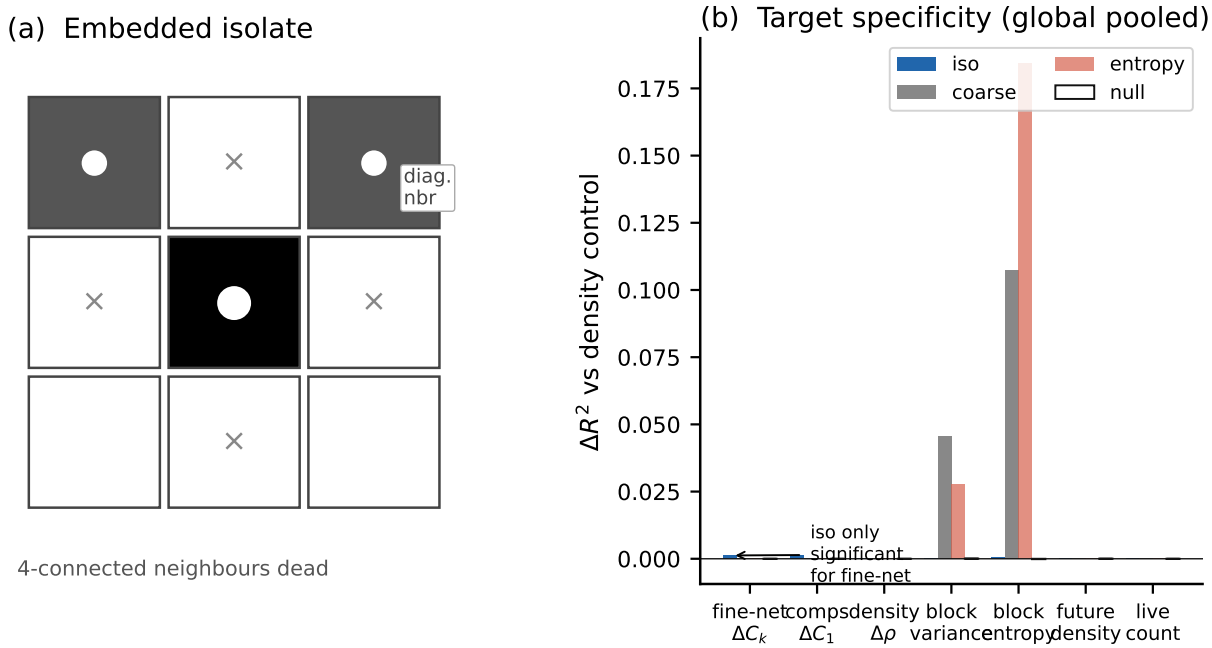

FIG. 1. **Result 1: structural definition and target-specific selection.** (a) An embedded isolate: an alive cell (filled circle) with all four orthogonal neighbours dead (×) and at least one diagonal neighbour alive (filled grey). This configuration places the cell at the most unstable locus of the local component topology. (b) ΔR^2 bars for iso_count and comparison predictors (coarse, entropy, null) on all seven prediction targets (global pooled). iso_count adds $\Delta R^2 = 0.0012$ for fine-net only; both nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not a density artefact.

$t = 0$ encodes the density of fragile bridge nodes, and this density alone—irrespective of which specific nodes change first—determines the aggregate response across all horizons. The mechanism thus acts in a statistically self-averaging way from the very first step.

V. RESULT 3 — TEMPORAL RESPONSE LAW

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

All 112 slopes are negative; all 112 slope confidence intervals are negative (frac_CI_negative = 1.00 at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$—a range of ≈ 0.09—while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains

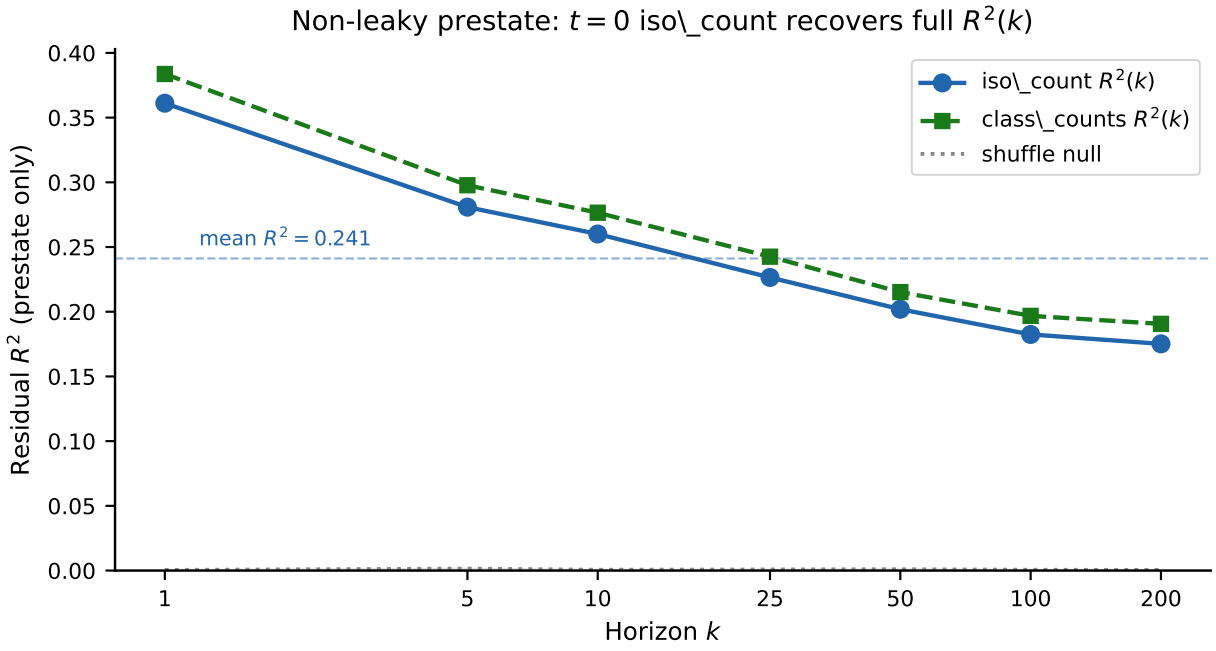

FIG. 2. **Result 2: non-leaky prestate recovery.** Using only $t = 0$ iso_count (no fate features, no local-window information), residual $R^2(k)$ stays above 0.175 at every horizon and averages 0.241 (dashed line). Adding class-count breakdown (green) gives marginal improvement. The grey band shows the shuffle null ($R^2 \approx 0.0008$). All temporal prediction is encoded in the $t = 0$ statistic.

well above the null at every horizon.

Figure 3 shows the slope and R^2 curves. The slope plateau from $k = 5$ to $k = 200$ indicates that the mechanism acts primarily in the first step: once the initial fragile nodes are resolved at $t = 1$, the remaining dynamics do not strongly modulate the iso_count signal. The CV profile confirms robustness: condition-level spread is small relative to the global mean, and GoL and HighLife track each other closely despite different neighbourhood rules.

Figure 4 provides the complete picture: a 16×7 heatmap of all condition-horizon slopes, every cell shaded negative. There are no outlier conditions or horizons.

VI. RESULT 4 — MECHANISM CARRIER

Having established *that* iso_count predicts fine-component change, we ask *how*: what actually happens at $t \to t + 1$ when embedded isolates reduce the future component count?

We decompose iso_count into one-step outcome categories: whether each isolate cell dies

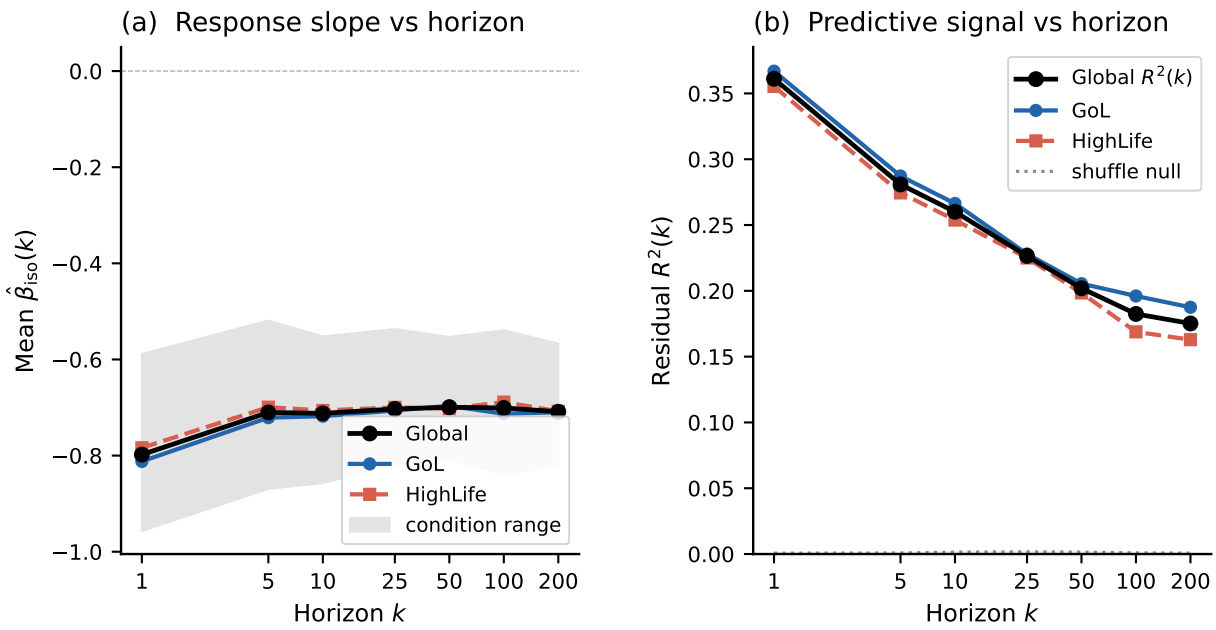

FIG. 3. **Result 3a: temporal response law slope and predictive signal.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (blue circles), HighLife (orange squares), and global mean (black); shaded region shows condition-level spread. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; grey band marks the shuffle null. Predictive power decays gracefully but remains far above null at all tested horizons.

TABLE I. Temporal response law: global summary (16 conditions, 1000 worlds each). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

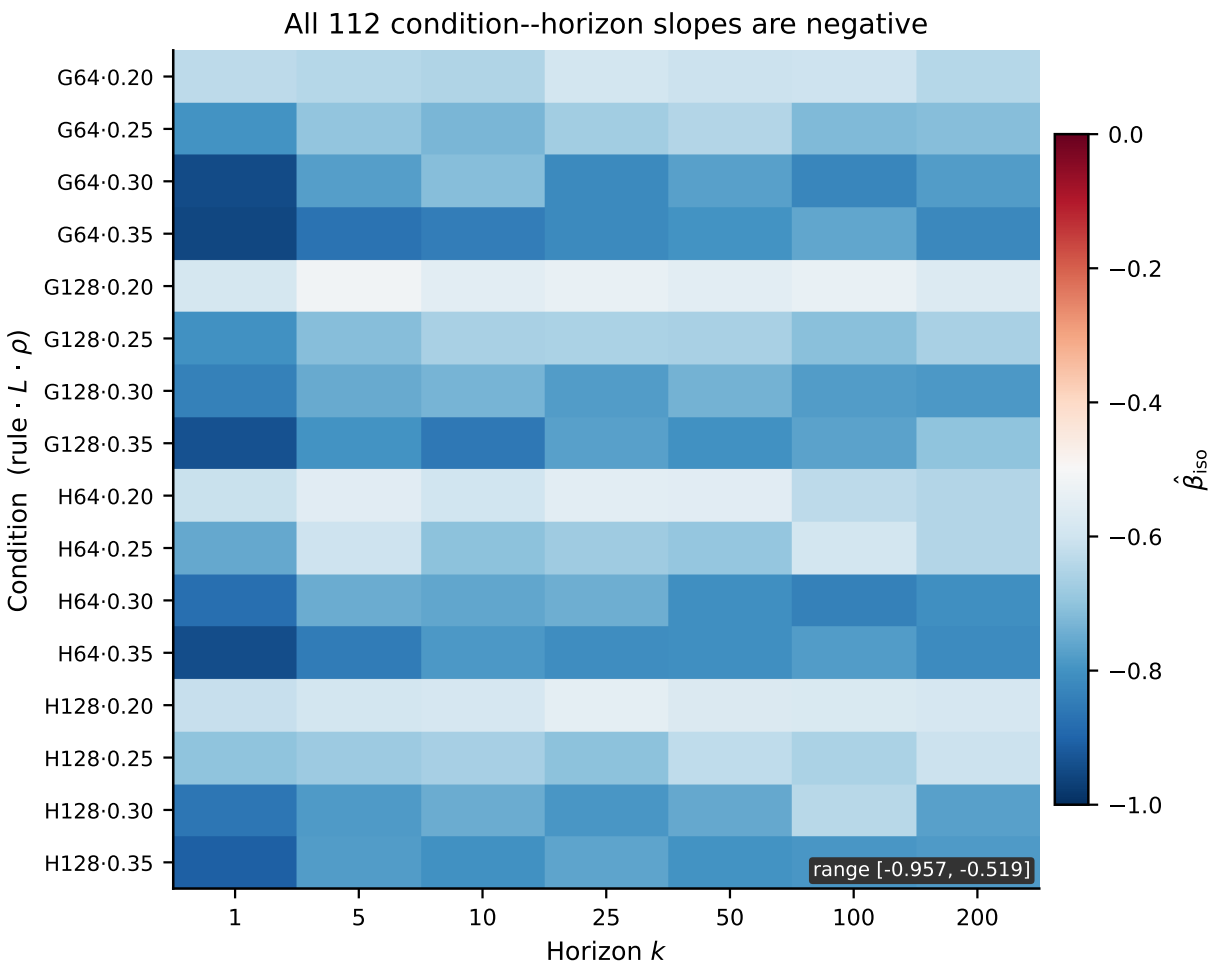

FIG. 4. **Result 3b: all 112 condition–horizon slopes are negative.** Full 16×7 heatmap of $\hat{\beta}_{\text{iso}}$ (RdBu_r, $[-1, 0]$). Rows are conditions (rule $\cdot$ L $\cdot$ ρ), columns are horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$. Every cell is negative, confirming the temporal response law without exception.

(iso_die), survives (iso_survive), survives and connects to a component (iso_survive_connected), or triggers nearby births at orthogonal (iso_orth_birth_any) or diagonal (iso_diag_birth_any) sites. We also compute a *local-window* feature: iso_local_window_loss_sum, the total component-count decrease within a 5×5 window centred on each isolate over one step. Cross-validated R^2 values for these features are shown in Fig. 5a.

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net

mechanism is *local component-context loss*, not simple cell death. Figure 5b shows the standardised slopes for each fate feature, revealing the partial cancellation between loss and gain.

The mechanism is not trivially explained by single-cell death statistics. Class-level analysis (Fig. 6) shows that even single-diagonal-neighbour isolates (the most common class, with death_frac = 1.0 in GoL) contribute comparable $p \times \overline{\Delta}$ as more complex classes, because their high prevalence compensates for their modest per-cell effect. The dominant process is the disappearance of fine-net connectivity in the local neighbourhood, which aggregates over many isolate classes.

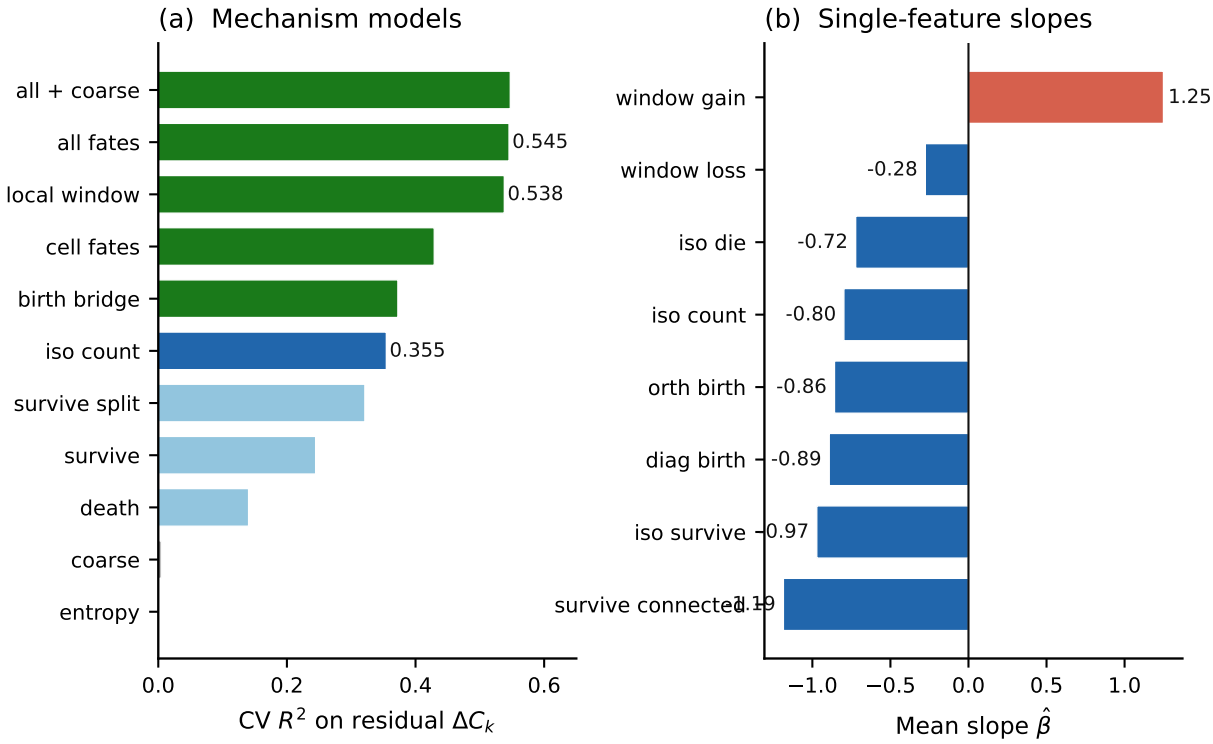

FIG. 5. **Result 4a: mechanism carrier decomposition.** (a) Cross-validated R^2 for each fate and mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features are negative, gain-related features are positive, revealing partial cancellation.

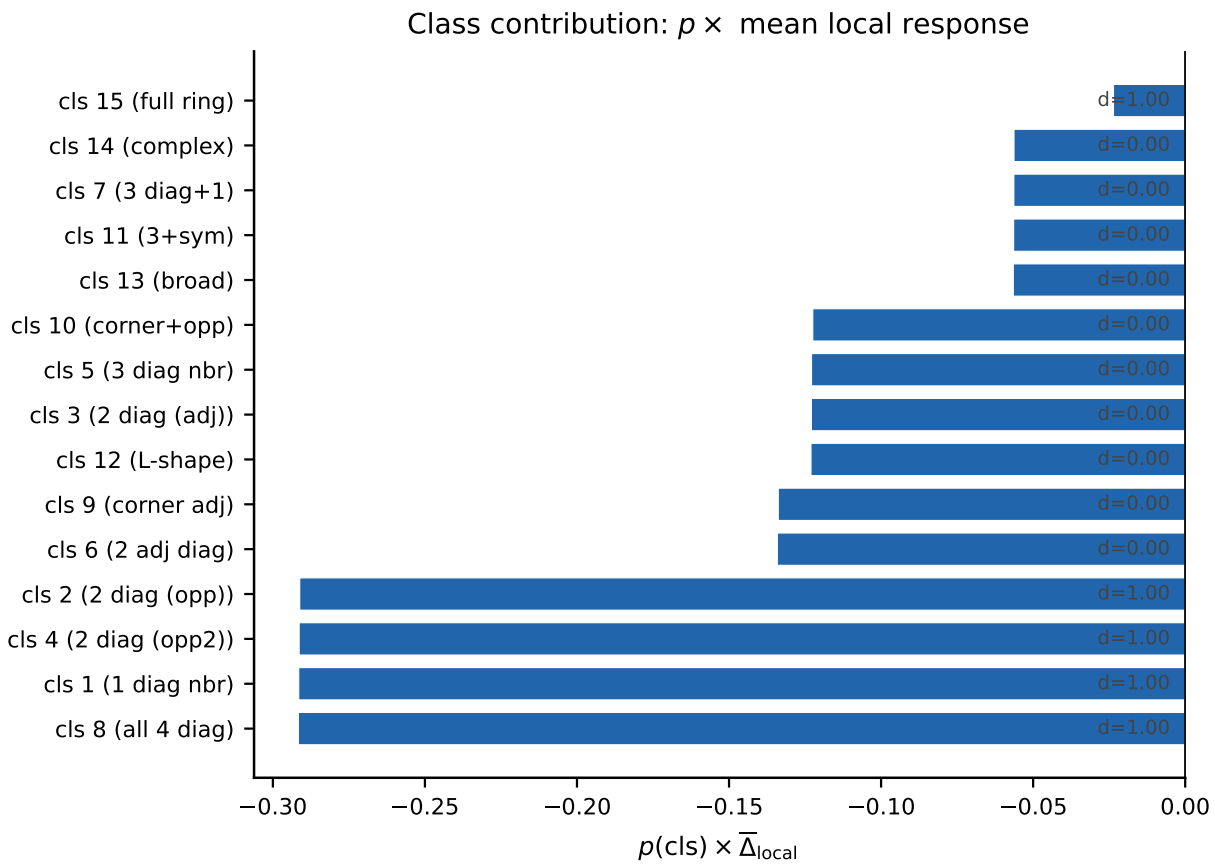

FIG. 6. **Result 4b: class-level contributions.** Horizontal bars show $p(\text{cls}) \times \overline{\Delta}_{\text{local}}$ for each diagonal-adjacency class. Annotations give the death fraction (d) from GoL transition-class statistics. All classes contribute negatively; single-diagonal-neighbour classes drive the bulk via their high prevalence.

VII. RESULT 5 — TRANSFER, AMPLITUDE LAW, AND TWO-LAYER STRUCTURE

The mechanism analysis establishes what the carrier is. Two separate questions remain: does it *generalise* to held-out conditions, and does the *scale* of the response follow a systematic pattern? These are distinct questions because generalisation operates on the standardised direction of the slope vector, while amplitude concerns its raw magnitude. A mechanism can transfer perfectly in direction while having condition-specific amplitude; conversely, amplitude can be predictable even when directions differ. We address them in turn.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

A. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). All four transfer axes pass (Fig. 7a). For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive $= 1.00$ on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$). The standardised mechanism direction is condition-invariant: whichever leave-one-out split is applied, a model trained on the remaining conditions predicts the held-out condition well.

B. Amplitude Law

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV $= 0.369$) but are predictable from (L, ρ). LOO R^2: full model $= 0.977$; size+ρ model $= 0.970$; size-only ≈ 0.802; rule-only $= -0.138$. Grid size L is the dominant predictor of amplitude; rule identity and density alone do not determine it.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L, \rho)\, \hat{\beta}_{\text{iso}}, z\,, \tag{1}$$

where $\hat{\beta}_{\text{iso}}, z \approx -0.50$ is the transferable standardised slope and $A(L, \rho)$ is a condition amplitude predictable with LOO $R^2 \approx 0.977$. The natural explanation is that L sets the number of components and hence the denominator of the relative change $\Delta C_k/C(0)$, while ρ controls the density of the specific structural class; together they set the amplitude scale that multiplies the universal standardised direction.

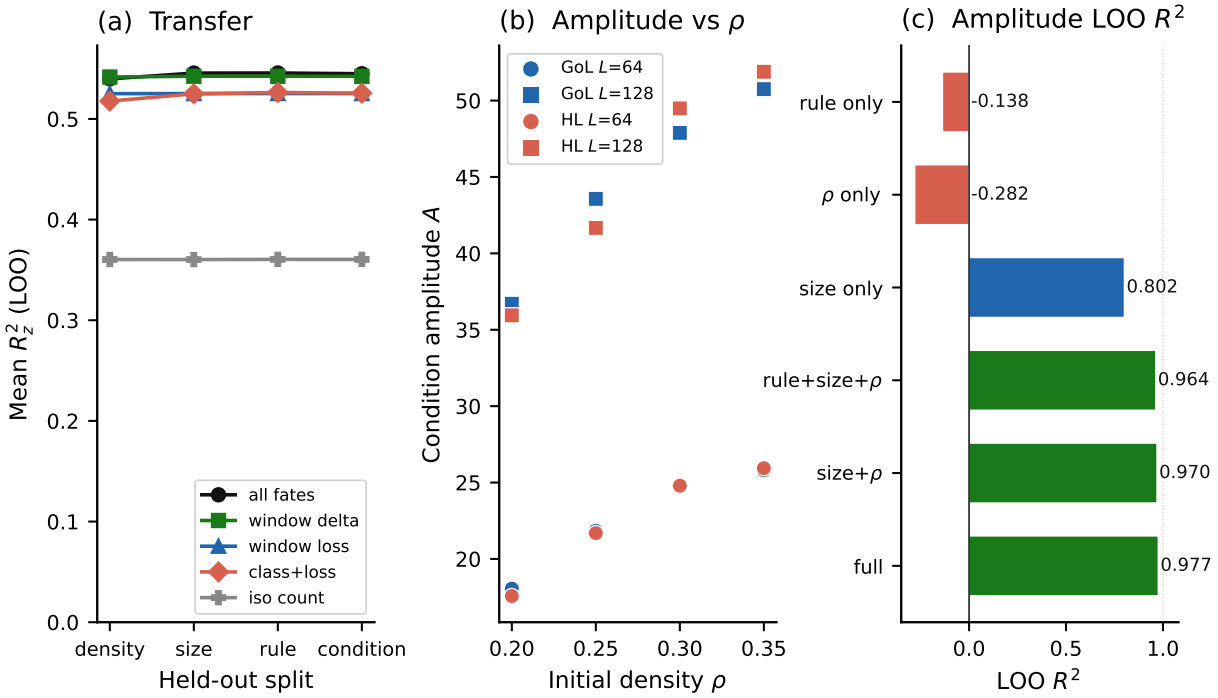

FIG. 7. **Result 5: transfer and amplitude law.** (a) Standardised transfer R_z^2 for five mechanism models across four leave-one-out axes; all lines stay positive. (b) Condition amplitude versus initial density ρ, coloured by grid size L and rule. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$, size+ρ alone reaches 0.970.

VIII. TASK-DIRECTION COHERENCE

The results above establish a strong empirical picture. We now ask whether there is a formal geometric property of the fine-net task family that explains *why* a single iso_count statistic can serve all seven tested horizons simultaneously.

The diagnostic proceeds as follows. For each prediction task (a specific horizon k or a specific prediction target), we fit a residual regression model, then normalize the coefficient vector to unit length in whitened feature space. This gives a unit "task-direction vector" for each task. We then compute the pairwise absolute cosines $|\cos\theta_{ij}|$ across tasks within the same family, and the fraction of the total spread variance captured by the leading singular value (rank-1 energy).

A family with $|\cos| \approx 1$ and rank-1 energy ≈ 1 lies essentially along a single direction

in description space: one feature set aligns all tasks simultaneously, which is precisely why a single statistic such as iso_count can serve the whole family. A family with $|\cos| \ll 1$ has geometrically dispersed tasks; no single feature can serve all of them.

For the fine-net horizon tasks, we find mean pairwise $|\cos| = 0.999$ and rank-1 energy $= 0.999$ (relative regret $= 0.001$). The fine-net family is essentially rank-1: a single description direction simultaneously aligns all k-step predictions. For the heterogeneous multi-target battery, by contrast, mean pairwise $|\cos| = 0.427$, rank-1 energy $= 0.532$, and mean relative regret $= 0.596$. No single description direction serves all heterogeneous targets.

This contrast provides a formal geometric account of the empirical results: iso_count works across all fine-net horizons because those tasks are nearly collinear in description space. The heterogeneous target family is not collinear, and indeed no single predictor adds $\Delta R^2 > 0.001$ for all targets simultaneously.

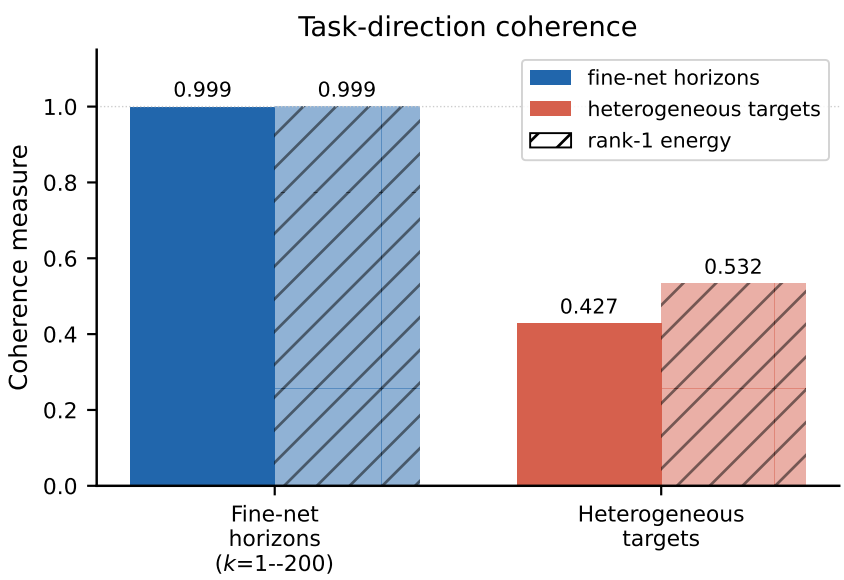

FIG. 8. **Task-direction coherence diagnostic.** Mean pairwise $|\cos\theta|$ (solid bars) and rank-1 energy (hatched bars) for fine-net horizon tasks (blue) and heterogeneous targets (orange). Fine-net horizons are near-perfectly aligned ($|\cos| = 0.999$, rank-1 energy $= 0.999$); heterogeneous targets are dispersed ($|\cos| = 0.427$, rank-1 energy $= 0.532$).

IX. DISCUSSION

What is earned. The five results form a coherent chain. *Selection (§III):* iso_count is a non-leaky target-specific predictor of fine-net component change, with $\Delta R^2 = 0.0012$ and incremental nulls near zero. *Prestate non-leakiness (§IV):* the full multi-horizon response is encoded in $t = 0$ iso_count alone; using fate information adds nothing substantial. *Temporal law (§V):* $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across $k \in \{1, \ldots, 200\}$, both rules, both grid sizes, and all four density levels; no exceptions in 112 condition-horizon combinations. *Mechanism (§VI):* local component-context loss (not simple cell death) is the one-step carrier; the local-window feature accounts for 0.538 of the residual variance, compared with 0.355 for iso_count alone. *Transfer and amplitude (§VII):* the standardised mechanism transfers across all leave-one-out axes while the raw amplitude has a clean two-layer structure—transferable direction, predictable scale—with LOO $R^2 = 0.977$. *Coherence (§VIII):* fine-net horizon tasks are geometrically coherent ($|\cos| = 0.999$); heterogeneous targets are not ($|\cos| = 0.427$).

The two-layer structure. The decomposition $\hat{\beta}_{\text{iso}}(\text{cond}) \approx A(L, \rho)\, \hat{\beta}_{\text{iso}}, z$ cleanly separates the universal mechanism from the condition-specific scale. This matters for applications: if a practitioner has access to a new (L, ρ) condition, they can predict the raw slope amplitude with LOO $R^2 \approx 0.977$ from (L, ρ) alone, and they can apply the standardised slope direction directly from the training pool. The two layers are independently portable.

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Embedded isolates as objects. Embedded isolated cells are not just a convenient statistic: they form a well-defined structural class at the topological boundary of connectivity. Their prevalence determines how much the fine-component count is "at risk" at the next step. The stability of $\hat{\beta}_{\text{iso}}(k)$ across horizons suggests that the risk resolves primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result. In this sense, iso_count is a *topological risk indicator* for fine-net change, rather than a simple density proxy.

X. CONCLUSION

The count of embedded isolated cells is the sufficient non-leaky prestate predictor of future fine-component change in Conway's Game of Life and HighLife. The response coefficient $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across all 7 tested horizons and all 16 conditions (2 rules $\times$ 2 sizes $\times$ 4 density levels), constituting an empirical temporal law. Both iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The one-step mechanism is local component-context loss, not simple cell death, and this standardised mechanism transfers across all leave-one-out axes. The raw response has a clean two-layer structure: a transferable standardised direction plus a predictable amplitude $A(L, \rho)$ (LOO $R^2 = 0.977$). Fine-net horizon tasks occupy a rank-1 coherent subspace of description space (mean pairwise $|\cos| = 0.999$), while heterogeneous prediction targets do not ($|\cos| = 0.427$), providing a formal geometric link to observer-dependent description theory.

ACKNOWLEDGMENTS

The author thanks the open-source communities behind NumPy, SciPy, scikit-learn, pandas, Matplotlib, and tqdm.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts of interest to disclose.

Author Contributions

K.B.: conceptualization, methodology, software, investigation, formal analysis, writing.

DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`.

REFERENCES

[1]J. C. Flack, "Coarse-graining as a downward causation mechanism," Philosophical Transactions of the Royal Society A **375**, 20160338 (2017).

[2]C. R. Shalizi and C. Moore, "What is a macrostate? Subjective observations and objective dynamics," arXiv preprint cond-mat/0303625 (2003).

[3]E. P. Hoel, L. Albantakis, and G. Tononi, "Quantifying causal emergence shows that macro can beat micro," Proceedings of the National Academy of Sciences **110**, 19790–19795 (2013).

[4]M. Gardner, "Mathematical games: The fantastic combinations of John Conway's new solitaire game "life"," Scientific American **223**, 120–123 (1970).

[5]C. G. Langton, "Computation at the edge of chaos: Phase transitions and emergent computation," Physica D: Nonlinear Phenomena **42**, 12–37 (1990).

[6]S. Wolfram, "Universality and complexity in cellular automata," Physica D: Nonlinear Phenomena **10**, 1–35 (1984).

[7]J. P. Crutchfield and K. Young, "Inferring statistical complexity," Physical Review Letters **63**, 105–108 (1989).

[8]A. Rupe and J. P. Crutchfield, "Local causal states and discrete coherent structures," Chaos **28**, 075312 (2018).

[9]N. Israeli and N. Goldenfeld, "Computational irreducibility and the predictability of complex physical systems," Physical Review Letters **92**, 074105 (2004).